Miaorun Wang (Dr. agr.)

Postdoctoral Researcher | Digital Agriculture

Rostock, Germany | miaorun.wang@uni-rostock.de | ORCID: 0000-0002-1401-9920

SCIENTIFIC PROFILE

Postdoctoral researcher with an interdisciplinary background in agricultural and environmental sciences, currently focusing on digital livestock farming. Experienced in statistical modelling, geospatial analysis, and sensor-based agricultural data across soil, crop, and livestock research.

ACADEMIC EXPERIENCE

Postdoctoral Associate, Project "iol.farm" – University of Rostock, Germany — Aug 2026 – Present

- Research in digital livestock farming within iol.farm, focusing on multisensor systems, sensor data analysis and validation of AI-based animal behaviour classification. Contributes to KI-TIERWOHL and AutoPasture on sensor- and AI-supported livestock monitoring.

Postdoctoral Associate (1-year position), DFG Project "Baltic TRANSCOAST" – University of Rostock, Germany — July 2024 – June 2025

- Analysed interdisciplinary environmental datasets on hydrology, soil properties and nutrient dynamics at the terrestrial–marine interface of coastal wetlands.

Postdoctoral Associate (1-year position), Project "MAPS" (Farmers' network: Multi-year analysis of crop protection strategies) – University of Rostock and xarvio® Digital Farming Solutions (BASF), Germany — May 2023 – June 2024

- Analysed long-term agricultural and environmental datasets, focusing on spatial and temporal variability of fungicide use in winter wheat and region-specific pesticide reduction strategies.

Doctoral Researcher, DFG Project "Baltic TRANSCOAST" – University of Rostock, Germany — Feb 2019 – Aug 2022

- Investigated hydrological processes, solute transport and spatial heterogeneity in peatland ecosystems using field experiments, statistical modelling and geospatial analysis.

EDUCATION

Dr.agr. in Soil Physics, University of Rostock, Germany — Feb 2019 – Jan 2024

M.Sc in Crop Production and Environment, University of Rostock, Germany — April 2017 – Jan 2019

B.agr. in Plant Protection , Nanjing Agricultural University, China — Sept 2011 – June 2015

RESEARCH METHODS & SKILLS

Data & (Geo)statistics: R, Python, QGIS; statistical modelling, geospatial analysis, time-series analysis

Digital Agriculture (on-going): sensor data analysis, reference measurements, camera-based validation, AI-supported animal monitoring (on-going)

Languages: Chinese (Native), English (C1), German (B2)

SELECTED PUBLICATION

1. Small-scale spatial variability of hydro-physical properties of natural and degraded peat soils — 2021

Wang, M., Liu, H., Lennartz, B.

Geoderma, 399(04):115123, https://doi.org/10.1016/j.geoderma.2021.115123

2. Temporal and spatial variability results of fungicide use in winter wheat (*Triticum aestivum* L.) recommend region-specific fungicide reduction target — 2025

Wang, M., Tackenberg, M., Andert, S. *Crop Protection*, 107281, https://doi.org/10.1016/j.cropro.2025.107281